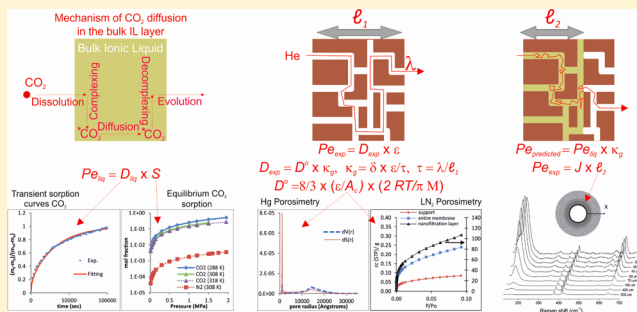


Alkyl-methylimidazolium Tricyanomethanide Ionic Liquids under Extreme Confinement onto Nanoporous Ceramic Membranes

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S Supporting Information

ABSTRACT: A method to predict the gas permeability of supported ionic liquid membranes (SILMs) was established, using as input the pore structure characteristics of asymmetric ceramic membrane supports and the physicochemical properties of the bulk ionic liquid (IL) phase. The method was applied to investigate the effect of IL nanoconfinement on the CO₂ and N₂ permeability/selectivity properties of novel SILMs developed on nanofiltration (NF) membranes employing for the first time the 1-ethyl-3-methylimidazolium and the 1-butyl-3-methylimidazolium tricyanomethanide ILs as pore modifiers. The selected ILs exhibit low viscosity, which allows for faster gas solvation rates and ease of synthesis/purification that makes them attractive for large-scale production. In parallel, the use of ceramic supports instead of polymeric ones presents the advantage of operation at elevated temperatures and pressures and offers the possibility to study the "real" permeability of the confined IL phase, avoiding additional contributions from the gas diffusion through the surrounding solid matrix. The developed SILMs exhibited enhanced CO₂ permeability together with high CO₂/N₂ separation capacity, though with distinct variations depending on the alkyl chain length of the 1-alkyl-3-methylimidazolium cation. Application of the developed methodology allowed discriminating the contribution of the NF pore structural characteristics on the SILM performance and unveiled the subtle interplay of diverse IL confinement effects on the gas permeability stemming from the specific layering of ion pairs on the nanoporous surface and the phase transition of the IL at room temperature, dictated by small variations of the IL cation size.



1. INTRODUCTION

The solubility and diffusion of several gases have already been studied in a high number of room-temperature ionic liquids (RTILs),^{1–20} and the results showed very promising CO₂ absorption capacity and CO₂/N₂ selectivity performance. However, one of the major drawbacks of RTILs is their high viscosity which leads to very slow CO₂ diffusion and makes their industrial scale application unfeasible. As a more convenient solution, immobilization of a thin layer of IL in sufficiently permeable membranes has already evidenced the possibility to overcome the problem of slow diffusivity and improve the efficiency of CO₂ removal from flue gas. Up to date RTILs have been immobilized in macroporous, polymeric^{21–32} supports and scarcely on inorganic nanoporous ones.^{33,34} Previous testing of polymer-based, supported ionic liquid membranes (SILMs) showed promising results with permeabilities/selectivities that were consistently above the upper bound of a Robeson plot for CO₂/N₂ separation.^{21,35–37}

The high performance of SILMs and the huge number of unexamined IL ion pairs have triggered researchers to develop methods for projecting the upper limits for SILM performance based on the physical chemistry of RTILs. These methodologies start with the assumption that gas transport through SILMs follows a dissolution/diffusion model

$$P = DS \quad (1)$$

where P is permeability; S is solubility (in moles per volume per partial pressure); and D is diffusivity. In a further step, models that describe the components, S and D , in RTILs are taken into account and compared against SILM data. Empirical models that can be mentioned are those of Camper,¹⁶ Kilaru,^{2,3} and Scovazzo^{38–40} for the prediction of gas solubility/selectivity and diffusivity in relation to the RTIL molar volume and viscosity. Eventually, the predicted permeability/selectivity performance

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of SILMs is compared to the experimental one, usually obtained via the time-lag^{15,41} technique, and the up-to-date results showed converging of experiment with prediction.⁴² A general conclusion of previous studies, encompassing all RTIL families that interact physically with CO₂, is that the critical properties affecting SILM performance are the molar volume and viscosity of the RTILs.⁴² Specifically, for CO₂/N₂ separation, decreasing the RTIL molar volume increases permeability/selectivity, and increasing viscosity decreases CO₂ permeability. On the basis of this, a number of future research foci are recommended including the study of anions (or cations) that will produce smaller molar volume RTILs and the combination of anions that produce smaller molar volume RTILs with functionalized cations, such as PEG and fluoroalkyls.

In this work, we have examined if the above-described methodologies and conclusions could be relevant to SILMs that have for the first time been developed on nanoporous (1 nm pore size) asymmetric ceramic membranes. Apart from the effect of IL nanoconfinement on the permeability/selectivity properties of the developed SILMs and the consequent deviation from our predictions, other effects such as those of the cation's alkyl chain length, the operation temperature, and the feed gas composition were also investigated. To complete the present study, new RTILs of low viscosity were synthesized and used to cast the SILMs. The employed RTILs were the 1-ethyl-3-methylimidazolium ([emim][TCM]) and 1-butyl-3-methylimidazolium tricyanomethanide ([bmim][TCM]). Both ILs exhibited CO₂/N₂ selectivity factors that exceeded the threshold of 100. The immobilization of the selected RTILs into the pore space of ceramic (1 nm pore size) membranes was achieved by means of pressure-assisted physical imbibition methods.

For developing the permeability prediction methodology, we considered the gas transport through a self-standing IL film to take place via a mechanism of dissolution/complexation/diffusion/decomplexation/evolution, and we calculated the permeability of the IL film (eq 1) by interpreting the results of gas sorption equilibrium (S) and transient states (D). We further implemented a detailed pore structure characterization for each layer of the asymmetric nanofiltration (NF) ceramic membrane to define the respective structure factors (κ_g). The κ_g characterizes the deviation of a pore system from ideality ($\kappa_g = 1$, for parallel noninterconnected pores of very smooth surface). The predicted permeability for the SILM was then derived as the product of the κ_g factor with the permeability of the self-standing IL film.

The proposed permeability prediction methodology in this work brings several advantages over the up-to-date applied procedures. First of all the effect of membrane pore structure is highly considered and expressed through its structure factor κ_g . Second, the gas diffusivity D and solubility S are defined experimentally rather than from empirical models. Moreover, the capacity of Raman spectroscopy to provide a detailed profile of the RTILs' penetration depth in the succeeding layers of the highly asymmetric membrane is exploited to accurately define both predicted and experimental permeability values.

2. EXPERIMENTAL SECTION

2.1. Chemicals and Materials. The ionic liquids used in this study were 1-ethyl-3-methylimidazolium tricyanomethanide ([emim][TCM]) (mass fraction purity >98%, with 800 ppm H₂O content) and 1-butyl-3-methylimidazolium tricyano-

methanide ([bmim][TCM]) (mass fraction purity >98%, with 150 ppm H₂O), which were synthesized by Iolitec on request and used as such. Commercial ceramic multilayered membranes encompassing microporous silica (NF) ultrathin selective layers have been used as porous supports with the purpose to develop the SILM systems. The membranes (monochannel monoliths of 15 cm length, 0.67 cm id, 1 cm od) with glazed ends (1.5 cm) were obtained from Inopor GmbH. The active surface of the NF layer was 26.3 cm², and its nominal pore size was 1 nm. The selective layer, located in the bore side of the monochannel monoliths, was supported by a macroporous α -alumina support (outer side) and two successive α -alumina layers of intermediate pore size. Permeability and absorption measurements were conducted with the following gases: He (99.999%), N₂ (99.999%), and CO₂ (99.998%).

2.2. Synthesis of the ILs. An amount of 1 mole of the corresponding imidazolium-based ionic liquid chloride (EMIM Cl or BMIM Cl, >98%, Iolitec) was dissolved in 1.5 L of dry dichloromethane. An amount of 1.05 mol of sodium tricyanomethide was added at once, and the reaction mixture was stirred for 48 h at RT. The mixture was then filtered through Celite, the mother liquor evaporated under reduced pressure, and the product dried under high vacuum for 24 h at 40 °C. Ionic chromatography showed Cl content of below 1%. Yields in both cases exceeded 90%.

2.3. SILM Casting. The development of the supported ionic liquid membranes was attained by applying a vacuum-assisted liquid infiltration technique. The membranes were accommodated into a specially designed tightly capped stainless steel module and initially degassed at 150 °C under vacuum. After lowering the temperature down to 30 °C, a certain volume of the bulk ionic liquid was suctioned through a throttle valve into the selective layer side to completely fill the bore space of the monolith within the module, while the annulus side was maintained under high vacuum. The IL was allowed for a couple of hours to wet the selective layer surface and be imbibed into its pores under the driving force of pressure difference, which was maintained at 8 bar for 2 h in the case of the membrane modified with the [bmim][TCM] and at 1 bar for 2 h in the case of the membrane modified with the [emim][TCM]. The sample codes are listed in Table 1.

Table 1. Codes Used for the Membranes Prepared and Examined in This Work^a

sample code	pore size	ionic liquid	modification method
M1	1 nm	-	untreated
M2	10 nm	-	untreated
SILM1	1 nm	[bmim][TCM]	combined vacuum- and pressure-assisted infiltration
SILM2	1 nm	[emim][TCM]	vacuum-assisted infiltration

^aThe chemical formulas of the infiltrated ionic liquids are C₁₀H₁₁N₅ and C₁₂H₁₅N₅.

2.4. Instrumentation. **2.4.1. Characterization.** Mercury porosimetry was performed with the Autoscan 25/60 (Quantachrome) mercury porosimeter. LN₂ porosimetry was performed with the Autosorb-1MP (Quantachrome) Nitrogen Porosimeter. SEM images were obtained using a JEOL JSM-7401F field emission gun-scanning electron microscope (FEG-SEM). Raman measurements were performed in backscattering configuration on a Renishaw inVia Reflex spectrometer using a

near-infrared (NIR) diode laser ($\lambda = 785$ nm) as the excitation source. Raman cross-section analysis was implemented on an encoded, feedback controlled, XYZ mapping sample stage that allowed recording line profiles with a step size of $0.1 \mu\text{m}$. Subtraction of the luminescence background on the Raman spectra was performed by cubic spline interpolation, while spectral deconvolution was carried out by nonlinear least-squares fitting of the Raman peaks to a mixture of Lorentzian and Gaussian line shapes. Differential scanning calorimetry (DSC) was obtained with the TA Instruments 2920 MDSC.

The ILs were characterized by measuring their density and viscosity within a temperature range of 278.15–363.15 K on an Anton Paar SVM 3000/G2 type stabinger, with an uncertainty of $\pm 0.0005 \text{ g/cm}^3$ for the density, $\pm 0.005 \text{ mPa}\cdot\text{s}$ for the viscosity, and $\pm 0.01 \text{ K}$ for the temperature. The physical properties of [emim][TCM] and [bmim][TCM] are presented in the respective section of the Supporting Information.

2.4.2. Experimental Setup for Permeability and Permeability Selectivity Evaluation. Permeability selectivity evaluation was performed in the Wicke–Kallenbach configuration, with a mixture of CO_2/N_2 in the range of 10–50% v/v CO_2 , sweeping the selective layer side of the SILMs (feed) at a pressure of 1 bar, and He sweeping their outer side (permeate), with a flow rate of $20 \text{ cm}^3\cdot\text{min}^{-1}$ at 1 bar. The supplied CO_2 and N_2 gases were mixed before entering the membrane cell. The feed, permeate, and retentate gas compositions were analyzed using an HP 5890 Series II gas chromatograph equipped with a stream selection valve, a gas sampling valve and TCD, and detector. The gas flow of each stream was controlled by electronic mass flow controllers (Bronkhorst F-200CV). The pressure at both sides of the SILMs was controlled by means of two electronic backpressure regulators (BPRs) (Bronkhorst P-702CV). Constant temperature was set via a PID controller and a Type K thermocouple in contact with the membrane cell. Mixed gas experiments were performed at 30, 60, and 80°C for all the developed SILMs.

Single gas permeability measurements were conducted at the same apparatus, applying a steady state technique (flow-through mode). The gas was supplied into the selective layer side of the membranes at a constant flow rate, predetermined by means of the electronic mass flow controller. By keeping the retentate outlet closed via a toggle valve, the supplied gas was forced to permeate through the empty pores (M1, M2) or the IL phase (SILM 1, SILM 2) and was withdrawn from the permeate side. Due to the mass transfer resistance of the membrane, a pressure difference was developed across the membrane sides, which was continuously monitored via a differential pressure transducer (ABB 2600T series). He permeability measurements were performed for the unmodified NF membranes at 100°C , whereas CO_2 permeability measurements were performed for the developed SILMs at 30, 60, and 80°C . The experimental setup and equations used to derive the permeability and selectivity factors are included in the Supporting Information.

2.4.3. Gas Sorption Test Unit. The CO_2 and N_2 gas solubility measurements were performed with a gravimetric microbalance (IGA 001, Hiden Analytical). The masses of the sample and counterweight pans, the hooks, the counterweight material, and the hang chains of the microbalance assembly were of the order of one to three hundreds of milligrams per item and were defined with an accuracy of $\pm 0.1\%$. The materials were appropriately selected to induce a symmetrical configuration to the balance setup to minimize buoyancy

effects. The microbalance had a $0.1 \mu\text{g}$ stable resolution. The amount of the bulk ionic liquid usually inserted was 90 mg. Before each measurement the bulk ionic liquid samples were degassed at 353 K and high vacuum (10^{-5} mbar). The densities of the gas bulk phase were calculated using the Benedict–Webb–Rubin equation of state for N_2 and CO_2 . The uncertainty for this equation lies in the order of 3%.⁴³ Regeneration conditions between successive measurements were 80°C , vacuum 10^{-5} mbar, until no mass loss.

3. RESULTS AND DISCUSSION

3.1. Determination of CO_2 and N_2 Solubility (S) in the New RTILs. Figure 1(a) presents the results of CO_2 and N_2

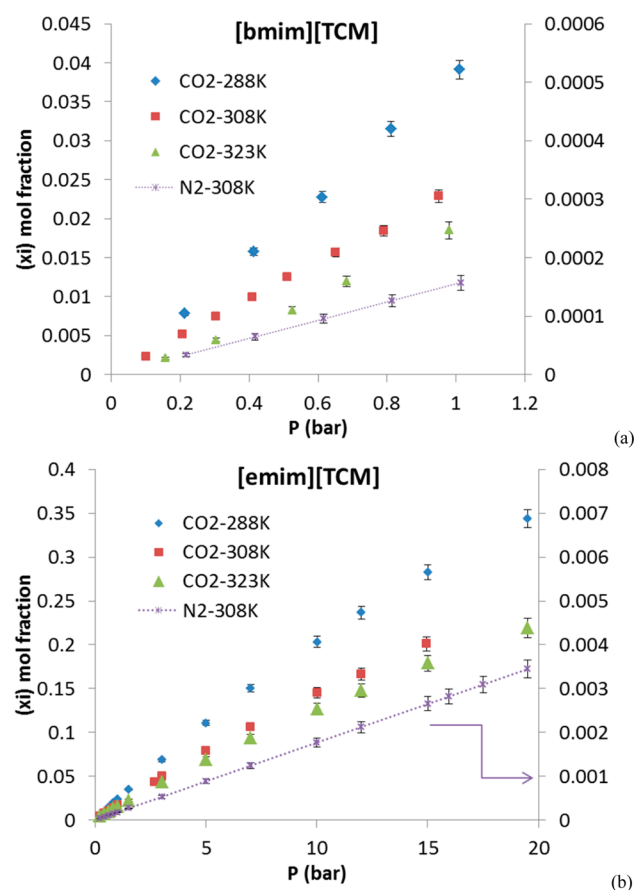


Figure 1. CO_2 and N_2 absorption isotherms. (a) [bmim][TCM], (b) [emim][TCM]. The error bars were extracted based on the uncertainty of the Benedict–Webb–Rubin equation in the calculation of the gas bulk phase density ($\pm 3\%$) and the change in molar liquid volume upon absorption of CO_2 which was based on a simple mole fraction average according to Shiflett et al.⁵⁰

gravimetric absorption measurements in the [bmim][TCM] at several temperatures up to the pressure of 0.1 MPa. Figure 1(b) depicts the respective results for [emim][TCM] up to 2 MPa. As observed, all absorption isotherms remained linear at the low-pressure region (<0.1 MPa), implying that CO_2 and N_2 solubility obey Henry's law for very dilute solutions of dissolved gases in liquids. CO_2 absorption has exothermic character since the Henry constant, $\bar{H}_{(\text{bar/mol fraction})}$, increased significantly with temperature, as shown in Table 2. $\bar{H}_{\text{CO}_2}$ for [emim][TCM] was larger than for [bmim][TCM] at all the examined temperatures, in agreement with previous reports suggesting that the

Table 2. Henry's Constants, CO₂/N₂ Ideal Sorption Selectivities, And Molar Volumes for the Studied RTILs at the Temperatures of the Absorption Isotherms

ionic liquid	T (K)	$\bar{H}_{\text{CO}_2}$ (bar/mol fraction)	$\bar{H}_{\text{N}_2}$ (bar/mol fraction)	S_{H} (–)	V_{m} (cm ³ ·mol ^{–1})
[emim][TCM]	288.15	41.65 ± 1.2	5658 ± 324	92 ± 5	211.03
	308.15	61.43 ± 2.4			213.52
	318.15	71.73 ± 3.4			214.90
[bmim][TCM]	288.15	25.27 ± 0.6	6467 ± 366	154 ± 10	217.40
	308.15	42.09 ± 1.6			220.19
	323.15	50.01 ± 2.3			222.34

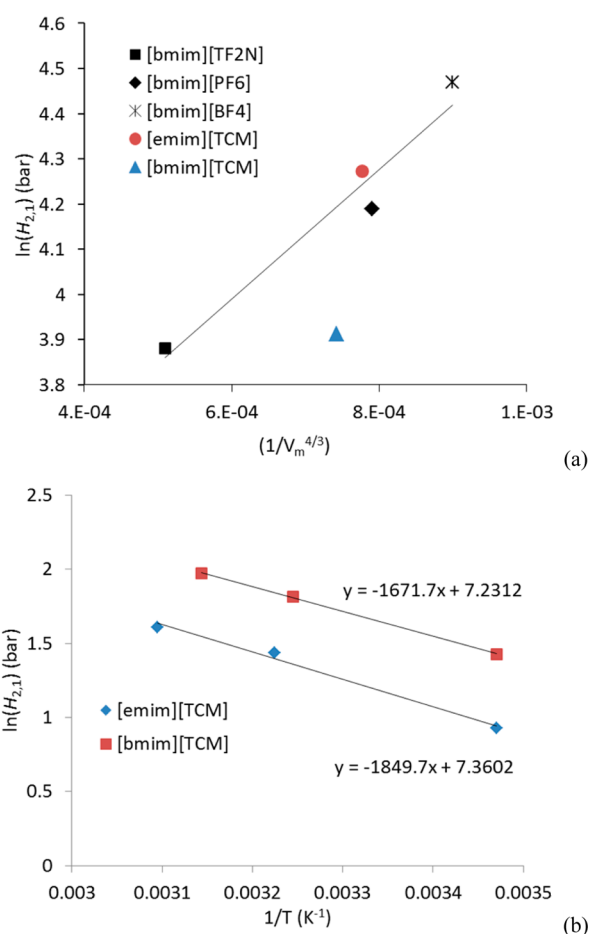
longer alkyl groups usually lead to slightly better CO₂ absorption capacity in the bulk ionic liquid.^{44,45} This trend has been ascribed to an increase in the liquid free volume, which is indicated by the decreasing density of ILs with increasing alkyl chain length (see Table S1 and Table S2, Supporting Information). The ideal CO₂/N₂ sorption selectivity (S_{H}) at the temperature of 308.15 K, derived from the ratio of the respective Henry's constants ($\bar{H}_{\text{CO}_2}/\bar{H}_{\text{N}_2}$), was 154 and 92 for the [bmim][TCM] and [emim][TCM], respectively, which is much higher than the S_{H} performances (<90) observed up to date for ILs that interact physically with CO₂^{18,46} (see Table S3 in the Supporting Information).

The CO₂ solubility results for [emim][TCM] and [bmim][TCM], were further examined and compared to those of other RTILs via a simplified version of the regular solution theory (RST)^{17,47} model, as proposed by Camper et al.⁸ In this approximation a modified version of the Kapustinskii equation was used to estimate the solubility parameters of imidazolium-based RTILs, and it was found that the solubility parameter of a RTIL (δ) (Hildebrand solubility parameter) is proportional to its molar volume, according to the relationship: $\delta \propto (1/V_{\text{m}}^{4/3})^{1/2}$. Accordingly, larger RTIL molar volumes correlate with lower $\bar{H}_{2,1}$ and therefore with larger gas solubility, S . Moreover, for [emim][X] and related RTILs, the Henry's constant at a given temperature can be predicted by their molar volume (or respectively by their pure liquid density). This correlation can be examined for the case of low-pressure CO₂ absorption in [bmim][TCM], [bmim][PF₆], [bmim][BF₄], and [bmim][Tf₂N] at 323.15 K and [emim][TCM] at 318.5 K. As shown in Figure 2(a), the values of $\ln \bar{H}_{2,1}$ (at 323.15 K) for [bmim][BF₄], [bmim][PF₆], and [bmim][Tf₂N] and for [emim][TCM] (at 318.15 K) correlate well with the pure RTIL molar volume ($R^2 = 0.9759$), in contrast to [bmim][TCM] (323.15 K). The negative deviation exhibited by [bmim][TCM] might reflect weak interactions between the cation and the anion that facilitate the solvation of CO₂ and better organization around the [TCM] anions.

Thermodynamic information on dissolved CO₂–solvent interactions was obtained by determining the enthalpy, Δh_{g} , and entropy, Δs_{g} , of absorption of CO₂ in both RTILs.

The Arrhenius plot of the Henry's constants for CO₂ absorption is depicted in Figure 2(b). As noticed in Table 3, [bmim][TCM] exhibited higher values for enthalpy and entropy of CO₂ absorption, compared to [emim][TCM], which are similar to the respective values for [bmim][PF₆]. The larger enthalpy of absorption implies the increased strength of interactions between the [bmim][TCM] and CO₂, whereas the larger entropy of absorption reveals a higher degree of ordering of dissolved CO₂ molecules (primarily around the TCM anions) that takes place upon dissolution of the gas.

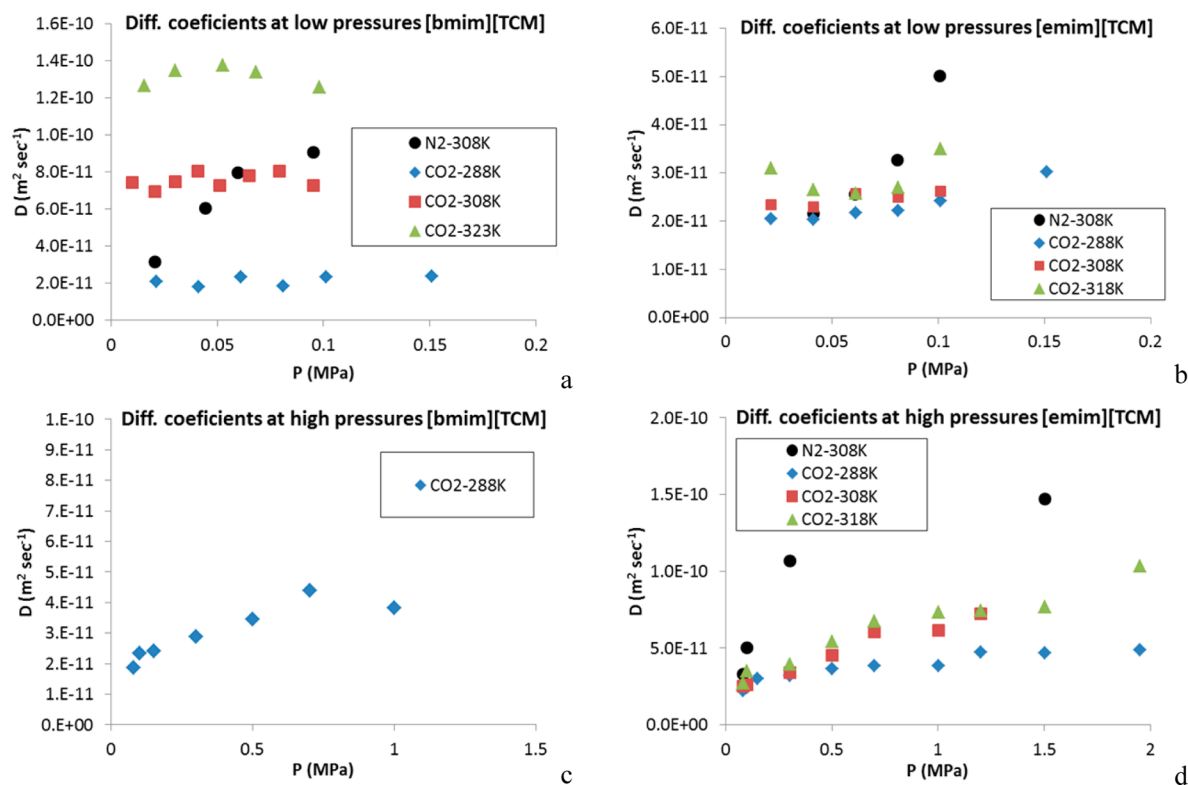
3.2. Transient Curves: Definition of N₂ and CO₂ Diffusivity (D) in the New RTILs. To investigate absorption

**Figure 2.** (a) Natural logarithm of Henry's constants for CO₂ absorption as a function of $V_{\text{m}}^{-4/3}$ for the [TCM]-bearing and non-[TCM] bearing RTILs. (b) Arrhenius plot of the Henry's constants for the studied RTILs.

kinetics, mass uptake data versus time were acquired from the sorption microbalance, for each equilibrium pressure and temperature and corrected for buoyancy. For the definition of the diffusion coefficient, D (m²·s^{–1}), of dissolved N₂ and CO₂ gases in the examined RTILs, the transient curves of absorption (mass uptake vs time) were fitted to the appropriate solution for the transient absorption curve, holding for diffusion rate controlled gas absorption in liquid films exhibiting one gas–liquid interface. The solution is derived from the transient diffusion equation according to a number of assumptions included in the Supporting Information.⁴⁸ The solution for the transient sorption curve is

Table 3. Enthalpies and Entropies of Dissolution for CO₂ in [emim][TCM] and [bmim][TCM] and Respective Data from the Literature

ionic liquid	Δh_{abs} (kJ mol ⁻¹)	Δs_{abs} (kJ mol ⁻¹)	temperature range (K)	ref.
[emim][TCM]	-13.9 ± 0.7	-46.02 ± 2.3	288.15–318.15	this work
[bmim][TCM]	-15.4 ± 0.5	-50.47 ± 1.5	288.15–323.15	this work
[emim][Tf ₂ N]	-14.0 ± 1		303.15–353.15	Camper et al. ¹¹
[bmim][PF ₆]	-16.1 ± 2.2	-53.2 ± 6.9	283.15–323.15	
[bmim][BF ₄]	-15.9 ± 1.3	-52.4 ± 4.3	283.15–323.15	Cadena et al. ⁷⁰
[emim][Tf ₂ N]	-14.2 ± 1.6	-46.9 ± 3	283.15–323.15	

**Figure 3.** Diffusion coefficients vs pressure: (a) [bmim][TCM] low-pressure region, (b) [emim][TCM] low-pressure region, (c) [bmim][TCM] high-pressure region, and (d) [emim][TCM] high-pressure region.

$$\frac{m_t}{m_\infty} = 1 - \frac{8}{\pi^2} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \exp\left[-\frac{D(2n+1)^2 \pi^2 t}{4l^2}\right] \quad (2)$$

where m_t is the total amount of gas, which has diffused into the liquid film at time t ; m_∞ is the total amount of gas that has diffused into the liquid after infinite time; and l (cm) is the liquid film thickness. The diffusion coefficients for the several pressure steps were calculated according to eq 2 for a liquid film thickness, l , of about 7×10^{-4} m, as calculated by the liquid mass and density and the diameter of the sample container used in the microbalance setup. Representative $(m_t - m_o)/(m_\infty - m_o)$ vs t plots for the studied RTILs are depicted in the Supporting Information, while the calculated diffusivities D (m²·s⁻¹) as a function of pressure are presented in Figure 3.

It can be observed that in the low-pressure region (0.01–0.1 MPa) the diffusion coefficient of CO₂ was independent of the dissolved phase concentration and thus could be treated as constant. In regard to N₂, the observed increasing trend can be attributed to inaccuracy of the measurement due to the very small amount of absorbed gas. Contrary, at higher pressures (0.1–2 MPa) the calculated D values of both gases were

increasing as the gas solubility increased as a consequence of the alteration of the IL viscosity due to the enhanced dissolution of the gas solute. It is mentioned that the diffusion coefficient of a gas species in the liquid is correlated with the solution viscosity, $D \propto \eta^{-n}$, where η is the solution viscosity and usually $0.5 < n < 1$ for various compounds.^{49,50} In addition, l (the depth of liquid in the container), which is a sensitive parameter to D , changes also when the gas dissolution is large. Thus, for the high-pressure range the analyzed diffusion constant must be regarded as an “apparent” one. However, it can be stated that the “apparent” D cannot differ significantly from the true value since the two error factors (η and l) caused by the large gas dissolution are counterbalancing (mutually canceling). It is also interesting to note that the CO₂ diffusivity in both ILs increases as a result of increasing temperature and the consequent decrease in their viscosity. The phenomenon is much more intense for [bmim][TCM], the viscosity of which drops by 45% in the temperature range of 293–323 K, when the respective drop observed for [emim][TCM] was 40%. Since η can be well correlated in a limited range of temperatures with the so-called Andrade equation,⁴⁹ ($\eta = Ae^{B/T}$), the temperature dependence of observed diffusion

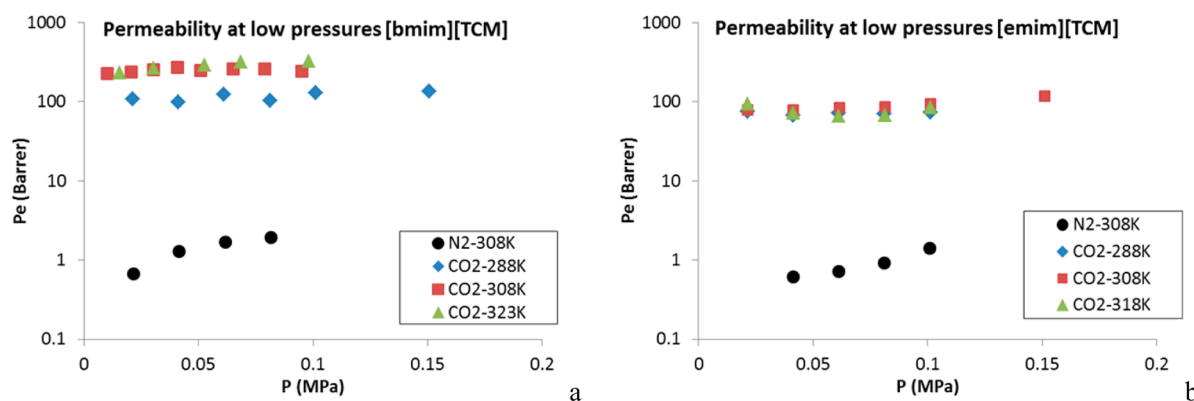


Figure 4. Predicted CO₂ and N₂ permeabilities of the developed membranes. (a) SILM 1, (b) SILM 2.

coefficients might be analyzed using an Arrhenius type of equation

$$\ln D = A - \frac{E_a}{RT} \quad (3)$$

Analyses using eq 3 showed that $E_a = 41 \text{ kJ} \cdot \text{mol}^{-1}$ and $E_a = 7.33 \text{ kJ} \cdot \text{mol}^{-1}$, for the binary systems CO₂ and [bmim][TCM] and CO₂ and [emim][TCM], respectively. It is interesting to note that these numerical values are outside the range of the free energy of activation in viscosity of hydrogen-bonded liquids ($12.6\text{--}25.1 \text{ kJ} \cdot \text{mol}^{-1}$).⁵¹

3.3. Permeability of a Self-Standing RTIL Film.

Introducing the as above-defined D and S values in eq 1, it was possible to calculate the CO₂ and N₂ permeability of the two RTILs in the form of thin films, at several temperatures and partial pressures. The diffusivity (D) was expressed in units of $\text{cm}^2 \cdot \text{s}^{-1}$ and the solubility (S) in units of $\text{cm}^3(\text{STP}) \cdot \text{cm}^{-3} \cdot (\text{cm}_{\text{Hg}})^{-1}$. Conversion to Barrer units was implemented by multiplying by the factor of 10^{-10} . The results are presented in Figure 4.

From an application point of view, we can suppose that the IL forms a semipermeable barrier of 0.07 cm thickness, acting as a self-standing liquid membrane. In this regard, the permeability values of 100–300 Barrer (Figure 4) when combined with the obtained selectivity S_{H} factors of above 100 generate coordinates that are far above the upper bound of a Robeson plot for CO₂/N₂ separation with membranes (see Supporting Information Figure S4). In a real application, however, the IL must be supported by a porous membrane. The IL confinement into the pores can considerably alter the gas diffusion properties of the liquid phase. For this reason a thorough investigation of the pore structure characteristics of the supporting material is required as described in the following paragraphs.

3.4. Characterization of the Ceramic NF Membranes.

The structure factor κ_g characterizes the deviation of a pore system from ideality ($\kappa_g = 1$, for parallel noninterconnected pores of very smooth surface) and is given by the expression

$$\kappa_g = \frac{\delta \varepsilon}{\tau} \quad (4)$$

where ε is the porosity; δ is the probability for a diffusing gas molecule to be randomly reflected after striking the pore wall ($\delta = 1$ for very smooth pore surface); and τ is the tortuosity which is a measure of the total length of an individual pore divided by the macroscopic membrane thickness.

To achieve the prediction of the gas permeability properties of a SILM, the permeability factor of a self-standing IL film (as calculated in the previous section) must be multiplied with the pore structure factor κ_g of the supporting ceramic membrane.

$$P_e = \kappa_g D S \quad (5)$$

What poses difficulties in the elucidation of κ_g is the highly asymmetric pore structure of the membranes used in this work for casting the SILMs. Depending on the possibility to control the imbibition process, the IL could occupy solely the pore system of the selective NF layer or be further infiltrated into the succeeding supporting layers. In the latter case more than one κ_g factor must be determined, and elucidation of the predicted permeability can be done by applying the model of resistances in series.

$$P_{e,\text{tot}} = \frac{P_{e,\text{NF}} \times P_{e,1}}{P_{e,\text{NF}} + P_{e,1}}, \quad \text{with } P_{e,\text{NF}} = \kappa_{g,\text{NF}} \times D \times S,$$

$$\text{and } P_{e,1} = \kappa_{g,1} \times D \times S \quad (6)$$

where $P_{e,\text{NF}}$, $\kappa_{g,\text{NF}}$ and $P_{e,1}$, $\kappa_{g,1}$ are the permeability coefficients and structure factors for the NF layer and the adjacent intermediate layers, respectively.

The usual way to define the κ_g of a membrane is by interpreting the results of porosimetry (LN₂ or Hg) and permeability of a nonadsorbable gas like He.

Thus, using the common definition of the permeability factor (eq 1)

$$P_{\text{exp}} = D_{\text{exp}} S = D_{\text{exp}} \varepsilon \quad (7)$$

where P_{exp} ($\text{cm}^2 \cdot \text{s}^{-1}$) and D_{exp} ($\text{cm}^2 \cdot \text{s}^{-1}$) are the experimentally defined He permeability and diffusion coefficient, respectively, and ε is the porosity ($\varepsilon \equiv S$ for the case of a nonadsorbable gas) and further applying the equation describing the Knudsen diffusion mechanism

$$D_{\text{exp}} = D^0 \kappa_g = \frac{8}{3} \kappa_g \frac{\varepsilon}{A_0} \left(\frac{2RT}{\pi M} \right)^{1/2} \quad (8)$$

where D^0 ($\text{cm}^2 \cdot \text{s}^{-1}$) is the thermodynamic or “ideal” diffusion coefficient for the case of parallel cylindrical pores of uniform diameter, R the gas constant, M the gas molecular weight, and A_0 ($\text{cm}^2 \cdot \text{cm}^{-3}$) the pore surface area of the membrane, it is possible to derive the κ_g .

It can be realized that the complexity of the asymmetric membrane support brings difficulties in the detailed elucidation of its pore structure. Indeed the above-described experimental

procedures must be applied to each layer separately. In the following we outline how it is possible to define the necessary parameters with the minimum possible requirement of experimentation. This is achieved by taking into account the geometrical characteristics of the tubular membrane, the skeleton density of the material constituting each layer, and the thickness of each layer as defined by SEM analysis.

3.4.1. Pore Structure Characterization of the Supporting Layers and the Thin NF Layer. The asymmetric morphology of the commercial α -alumina/silica tubular membrane (M1) was investigated via FE-SEM microscopy. The entire membrane consisted of an ultrathin (1.5 μm) NF, silica selective layer (Figure 5a), two intermediate α - Al_2O_3 layers of thickness $l_{il,1} = 39.2 \mu\text{m}$ and $l_{il,2} = 25.4 \mu\text{m}$, respectively (Figure 5b), and a coarse macroporous α - Al_2O_3 support located at the shell side of the tube. The respective mean pore sizes were 0.21, 0.6, and 2.84 μm as defined by the pore size distribution plot ($D_V(r)$ vs r) of mercury (Hg) porosimetry (Figure 5b). More important,

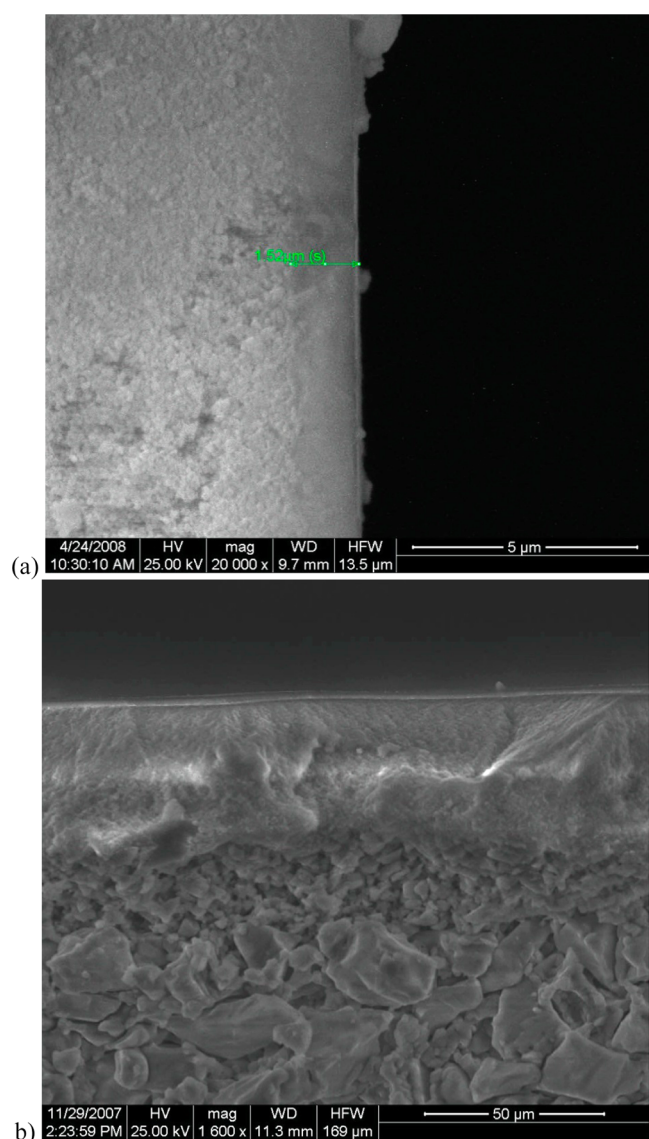


Figure 5. SEM micrographs of the cross-section of a commercial composite nanofiltration membrane. (a) Silica separation layer and (b) intermediate support layers plus outer coarse support.

however, was the definition of the porosity (ϵ) and surface area (A_o) of each layer, the A_o expressed in units of $\text{cm}^2 \cdot \text{cm}^{-3}$, to be directly applied in eqs 7 and 8 for the calculation of the respective κ_g . The elucidation of the pore structure of each of the three supporting layers (except from the top NF) was achieved by means of one experiment of Hg intrusion up to the pressure of 60 000 psig. By elaborating the mercury intrusion data (Figure 6a), the pore size distribution $D_V(r)$ ($\text{cm}^3 \cdot \text{g}^{-1} \cdot \text{\AA}^{-1}$)

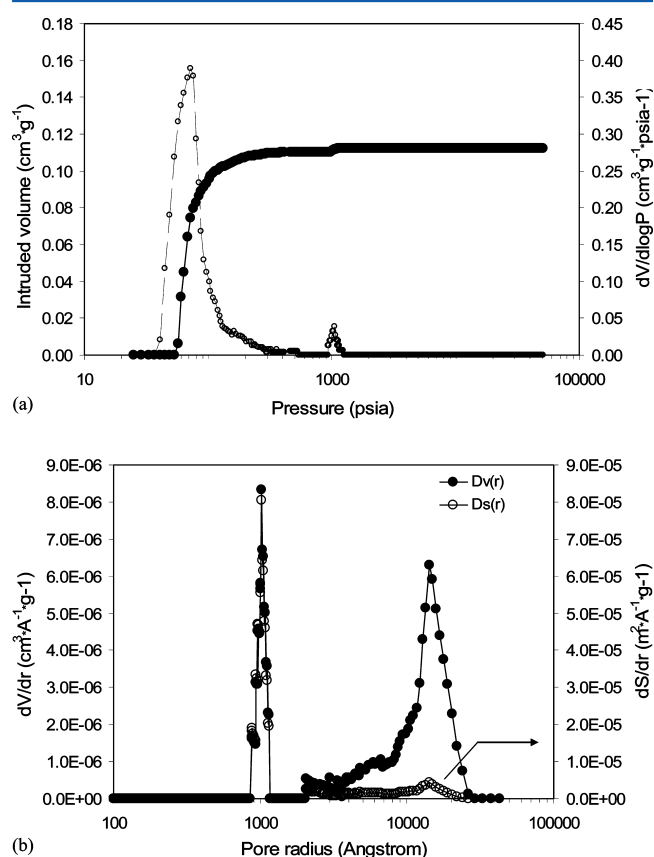


Figure 6. (a) Mercury intruded volume and $dV/d \log P$ vs applied pressure curves. (b) Pore size distribution and pore surface distribution of the support.

and pore surface distribution $D_S(r)$ ($\text{m}^2 \cdot \text{g}^{-1} \cdot \text{\AA}^{-1}$) of the entire membrane (M1) (Figure 6b) were defined according to the following equations

$$|D_V(r)| = \frac{3.44738 \times 10^{-4} P^2}{\gamma \cdot \cos \theta} \cdot \frac{dV}{dP} \quad (9)$$

$$|D_S(r)| = \frac{D_V(r)}{r} \quad (10)$$

where P is pressure in psia; θ is the contact angle for mercury (140°); γ is the surface tension of mercury ($480 \text{ erg} \cdot \text{cm}^{-2}$); r is the pore radius in $\text{\AA}$; and V is intruded volume (pore volume) in cm^3 .

Definition of the pore volume and surface area of each layer was achieved via integration of the above equations over the specific pore radius range which corresponded to each of the layers (Figure 6b). However, these results were useless since they are normalized over the entire mass of the membrane support. Therefore, it was necessary to calculate the absolute pore volume V_{pi} (cm^3) and surface area S_{pi} (m^2) of each layer in

Table 4. Structural Parameters of the Commercial Membranes Support

membrane layers	l (μm)	ρ_s ($\text{g}\cdot\text{cm}^{-3}$)	S_l ($\text{m}^2\cdot\text{g}^{-1}_{\text{layer}}$)	A_o ($\text{cm}^2\cdot\text{cm}^{-3}_{\text{layer}}$)	ε (—)	V_i ($\text{cm}^3\cdot\text{g}^{-1}_{\text{layer}}$)
selective NF layer	1.5	2.2	360	9×10^6	0.22	0.128
intermediate 1	39.2	3.96	2.31	7.42×10^4	0.188	0.064
intermediate 2	25.4	3.96	4.42	6.074×10^4	0.653	0.52
outer layer	1600	3.96	0.15	0.493×10^4	0.170	0.053

the specific sample of membrane (of length L) used in the experiment, by multiplying the normalized results with the mass of the membrane sample. Furthermore, the geometrical volume V_{gi} of each layer was defined using the inner tube diameter, d_{in} , outer tube diameter, d_o , thickness of the intermediate layers 1 and 2, $l_{il,1}$ and $l_{il,2}$, respectively, and length of the membrane sample, L , as inputs in the following equations

intermediate layer 1:

$$V_{g,1} = \frac{\pi L}{4} [(d_{in} + 2l_{il,1})^2 - d_{in}^2] \quad (11)$$

intermediate layer 2:

$$V_{g,2} = \frac{\pi L}{4} \{ [d_{in} + 2(l_{il,1} + l_{il,2})]^2 - (d_{in} + 2l_{il,1})^2 \} \quad (12)$$

outer layer:

$$V_{g,3} = \frac{\pi L}{4} \{ d_o^2 - [d_{in} + 2(l_{il,1} + l_{il,2})]^2 \} \quad (13)$$

For each individual supporting layer, the pore surface area, A_o (in $\text{cm}^2\cdot\text{cm}^{-3}$), was then derived as $A_{oi} = S_{pi}/V_{gi}$ and the porosity ε as $\varepsilon = V_{pi}/V_{gi}$.

The elucidation of the pore structure of the top NF layer was achieved by means of two LN_2 (77 K) porosimetry experiments which were performed on one piece of membrane without the selective layer (support (M2)) and on one piece of the entire NF membrane, respectively (M1). The support isotherm was then subtracted from that for the entire membrane at each relative pressure as follows

$$V_{NF} \left(\frac{cc(STP)}{g} \right) = \frac{V_{mem}(cc(STP)) - V_{sup} \left(\frac{cc(STP)}{g} \right) x M_{mem}(g) x R_{sup}}{M_{mem}(g) x R_{NF}} \quad (14)$$

where V_{NF} is the standard gas volume per mass adsorbed by the nanofiltration layer, V_{mem} the absolute standard gas volume adsorbed by the entire membrane, V_{sup} the standard gas volume per mass adsorbed by the support, M_{mem} the mass of the entire membrane, R_{sup} the mass ratio of the support, R_{NF} the mass ratio of the nanofiltration layer later defined from the geometrical dimensions and the solid density (d) of the materials consisting of the NF layer (silica, $d = 2.2 \text{ g/cm}^3$, thickness $1.5 \mu\text{m}$) and the support (alumina, $d = 3.96 \text{ g/cm}^3$, $id = 6.7 \text{ mm}$, $od = 10 \text{ mm}$).

The derived NF layer isotherm was interpreted with the DR approximation to define the microporosity (ε) and the micropore surface area (A_o). Values of the pore structure parameters of all the layers constituting membrane M1 are listed in Table 4.

3.4.2. He Permeability Definition of the Structure Factor κ_g . Helium permeability measurements were performed at 100°C to ensure that Knudsen type was the dominant diffusion mechanism through the membranes and avoid the effect of surface diffusion. Raman analysis (see section 3.5) defined imbibition of the RTILs up to a depth up to 40 and $25 \mu\text{m}$ for SILM 1 and SILM 2, respectively. Consequently, to be able to predict the permeability of SILM 1 and SILM 2, it was adequate to define the κ_g factors of solely the NF layer ($1.5 \mu\text{m}$) and the intermediate layer 1 ($39.2 \mu\text{m}$). Therefore, He permeability measurements were conducted on membrane (M1) in which gas transport is totally controlled by the NF selective layer and on membrane (M2), in which gas transport is controlled by the intermediate layer 1. The He permeability results together with the defined structure factors (eq 8) are presented in Table 5.

Table 5. Experimental Values of He Permeability (P_{exp}) for Membranes M1 and M2 and the Respective Structure Factors of the NF Layer and the Intermediate Layer 1

membrane	He permeability experimental ($\times 10^3$) ($\text{cm}^2\cdot\text{s}^{-1}$)	D_{exp} ($\times 10^3$) ($\text{cm}^2\cdot\text{s}^{-1}$)	D^o ($\text{cm}^2\cdot\text{s}^{-1}$)	κ_g (—)
M1/ $\kappa_{g,NF}$	0.00158	0.0072	0.00459	0.00156
M2/ $\kappa_{g,1}$	1.39	7.393	0.587	0.0126

3.5. Raman Spectroscopy. Figure 7 compares the Raman spectrum on the surface of the NF membranes after adsorption of [emim]/[bmim]/[TCM] with the corresponding ones of the

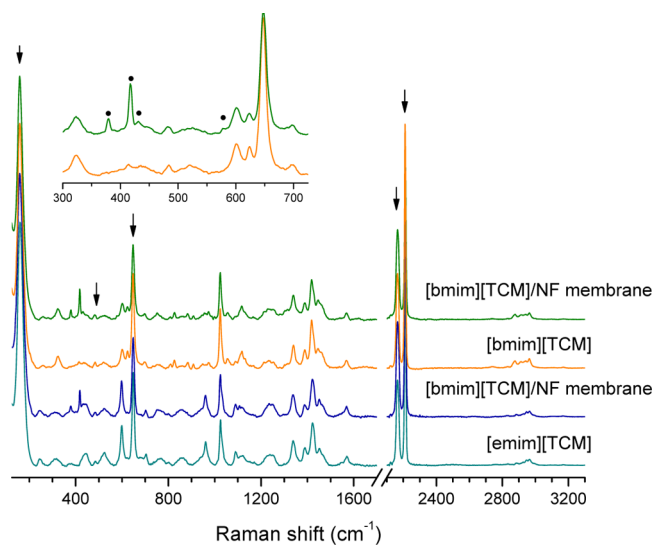


Figure 7. Raman spectra of the neat [emim][TCM] and [bmim][TCM] ILs in the liquid state and upon adsorption on the surface of the NF membranes. Arrows indicate the Raman modes of the $[\text{C}(\text{CN})_3]^-$ anion. The inset shows in detail the Raman spectra in the region of $300\text{--}725 \text{ cm}^{-1}$ with dots depicting the $\alpha\text{-Al}_2\text{O}_3$ Raman modes.

neat ILs in the liquid state. The adsorption of the ionic liquids on the membrane surface was thereby directly verified by the distinctive [emim]/[bmim] cation and tricyanomethanide anion Raman bands on the membrane support. All the characteristic Raman bands of [emim]⁺ and [bmim]⁺ cations were observed in the liquid state of the pure ILs, in agreement with previous reports for the corresponding 1-alkyl-3-methylimidazolium ILs with different anions.^{52–54} In particular, [emim][TCM] presented a broad band at $\sim 443\text{ cm}^{-1}$ related to the contribution of two distinct modes arising from the nonplanar (430 cm^{-1}) and planar (448 cm^{-1}) 1-ethyl-3-methylimidazolium conformers, respectively.^{55,56} Likewise, the coexistence of four [bmim]⁺ conformation states was identified by the observation of the weak but well-defined Raman modes at 809, 827, 884, and 908 cm^{-1} , assigned to the corresponding combinations of the gauche (G) and anti (A) [bmim]⁺ conformers (GG, GA, AG, and AA).⁵⁷ Conformational isomerism in [bmim][TCM] was further evidenced by the presence of a characteristic band at 324 cm^{-1} stemming from both GA and AA conformers, together with two discrete bands at 601 and 624 cm^{-1} , associated with the contributions of the GG/GA and AG/AA groups, respectively.^{57,58} The Raman vibrational modes of the [TCM][−] anion were observed at 158, 483, 648, 2165, and 2211 cm^{-1} , corresponding to the in-plane CCN bending mode, the out-of-plane bending, the symmetric CC stretching, and the antisymmetric and symmetric CN stretching vibrations, respectively.^{59–61}

Adsorption of both [emim][TCM] and [bmim][TCM] on the NF membranes resulted in negligible variations of the cation Raman modes, while a series of narrow bands emerged at 377, 417, 429, 448, 577, and 752 cm^{-1} that can be identified with the Raman modes of the $\alpha\text{-Al}_2\text{O}_3$ inner membrane layers^{62,63} (no major contribution of the thin silica separation layer could be traced). Similarly, the [TCM][−] Raman modes were slightly affected by the adsorption of the ionic liquid on the membrane (only a rather weak broadening and shift to higher frequencies by 1 cm^{-1} could be traced), most likely due to the relatively large scattering volume probed by the laser beam extending several micrometers below the NF silica layer. In that case, the contribution of nonspecifically adsorbed IL molecules from the intermediate $\alpha\text{-Al}_2\text{O}_3$ membrane layers dominated the Raman spectrum, thus effectively masking the IL Raman response from the NF layer, where signatures of interaction and confinement effects on the ILs structure would be anticipated.

To determine the ILs' adsorption profile across the membrane thickness, Raman analysis was subsequently performed along a membrane cross-section, as exemplified for [bmim][TCM] in Figure 8a. Moving away from the internal membrane surface resulted in a characteristic variation of the [bmim][TCM] Raman intensity relative to that of $\alpha\text{-Al}_2\text{O}_3$, which allowed quantification of the ionic liquid infiltration depth in the membrane support. Figure 8b shows the thickness (x) dependence of the integrated intensity ratio of the most intense [TCM][−] in-plane bending mode at 158 cm^{-1} relative to that of the 416 cm^{-1} mode of $\alpha\text{-Al}_2\text{O}_3$. The intensity ratio I_{158}/I_{416} for [bmim][TCM] presented a rather weak variation up to ca. $40\text{ }\mu\text{m}$, matching closely the thickness of the NF silica layer ($1.5\text{ }\mu\text{m}$) together with the first $\alpha\text{-Al}_2\text{O}_3$ intermediate layer ($39\text{ }\mu\text{m}$) with pore size of 210 nm . A rapid decrease of the I_{158}/I_{416} intensity ratio was further observed at $x > 45\text{ }\mu\text{m}$, reaching a plateau in the range of $50\text{--}65\text{ }\mu\text{m}$, which indicates a substantial reduction of the ionic liquid adsorption on the significantly

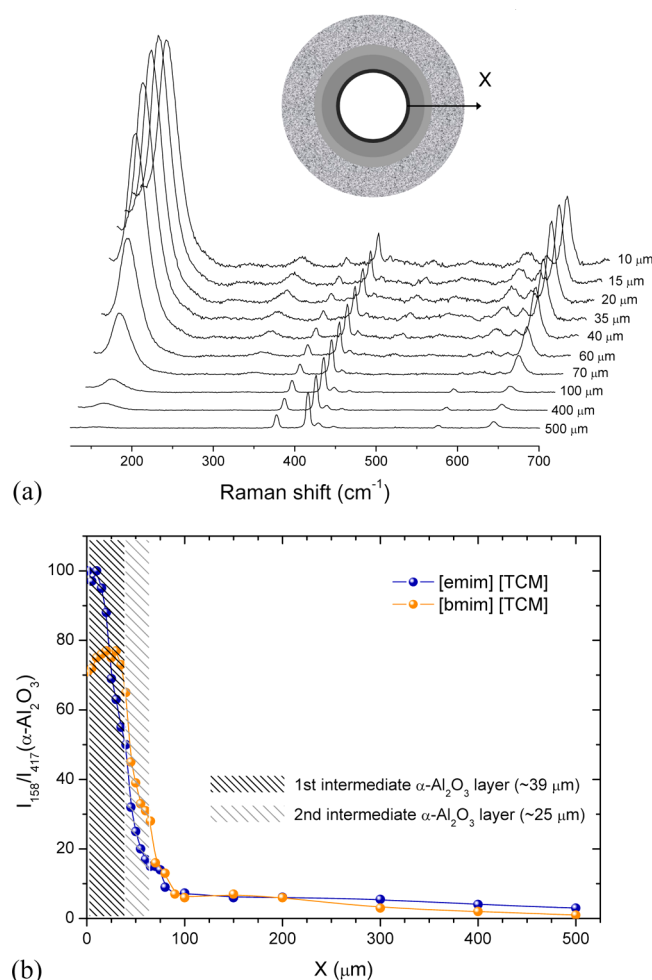


Figure 8. (a) Raman spectra on a cross-section of the [bmim][TCM] modified NF membrane at various distances from the internal surface. (b) Thickness (x) dependence of the integrated intensity ratio I_{158}/I_{416} along the membrane cross section. Shaded areas depict the first and second $\alpha\text{-Al}_2\text{O}_3$ intermediate layers of 39 and $25\text{ }\mu\text{m}$, respectively.

larger pores (600 nm) of the second $\alpha\text{-Al}_2\text{O}_3$ intermediate layer ($25.4\text{ }\mu\text{m}$). Consequently, it can be deduced that no pore filling by the IL phase occurred, in contrast to the nanofiltration and the first intermediate layers. Further decrease of the intensity ratio was detected up to $100\text{ }\mu\text{m}$ above which I_{158}/I_{416} degraded smoothly toward zero at $\sim 500\text{ }\mu\text{m}$, reflecting the diminished adsorption of the ionic liquid in the macroporous $\alpha\text{-Al}_2\text{O}_3$ support (pore size of $2.84\text{ }\mu\text{m}$). In the case of [emim][TCM], an even higher I_{158}/I_{416} ratio was recorded up to ca. $10\text{ }\mu\text{m}$ compared to [bmim][TCM], indicative of a higher concentration of adsorbed IL molecules in the membrane pores due to their relatively smaller size. Moreover, the intensity ratio I_{158}/I_{416} decreased continuously at $x > 20\text{ }\mu\text{m}$ through the first and second $\alpha\text{-Al}_2\text{O}_3$ intermediate layers up to $100\text{ }\mu\text{m}$, followed by a gradual suppression up to $x \sim 500\text{ }\mu\text{m}$ in the macroporous $\alpha\text{-Al}_2\text{O}_3$ layer.

3.6. Comparison between the Predicted and Experimental CO_2 and N_2 Permeability and CO_2/N_2 Selectivity. The experimentally derived CO_2 and N_2 permeability factors and the CO_2/N_2 selectivities at 30 and $60\text{ }^\circ\text{C}$ are presented in Figure 9, together with the predictions. The predicted permeabilities were defined with the use of eq 6, and the ideal CO_2/N_2 selectivities were calculated as the ratio of the

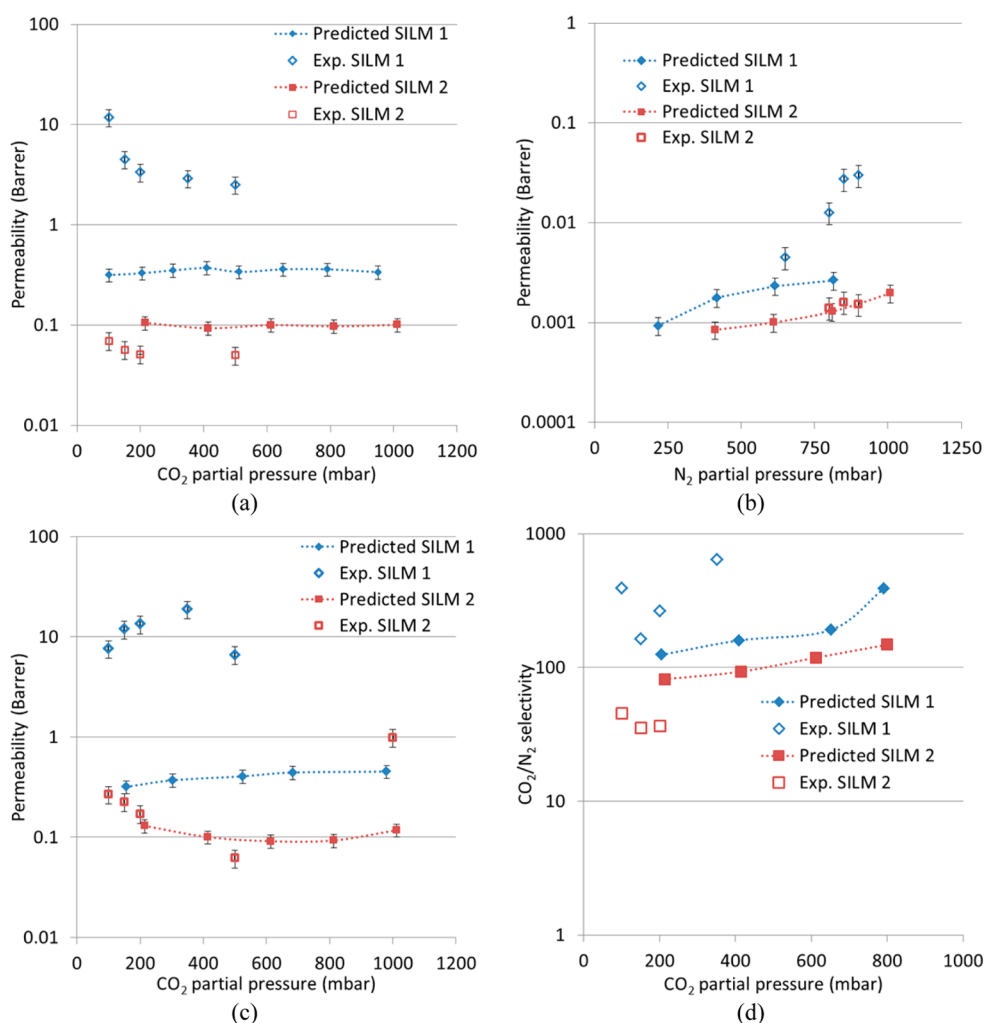


Figure 9. (a) Experimental and predicted permeabilities of gases at 30 °C as a function of their partial pressure in the feed. (a) CO₂ and (b) N₂. (c) Experimental and predicted permeabilities of CO₂ at 60 °C. (d) Experimental and predicted CO₂/N₂ selectivities at 30 °C.

predicted CO₂ and N₂ permeability values. It can be realized that our experimental results deviate considerably from the predictions. This comes into contradiction with the usually observed behavior of SILMs cast on polymeric macroporous supports and anodized aluminum oxide (AAO) disks.^{42,64} More specific, in polymer-based SILMs, mixed gas selectivities are approximately equal to the single gas or ideal selectivities. This issue confirms the statement that solubility selectivity dominates the permeability selectivity in SILMs, with the diffusion selectivity not playing an important role.³⁵ In the case of ceramic supports such as AAOs with pore sizes of 20 and 100 nm, experimental permeability was larger than the one predicted from the fundamental property values of ILs. Especially, for most of the examined ILs, predicted and experimental values showed insignificant deviation, whereas for ILs such as the [C₄C₄mim][Tf₂N] the values differed by a factor of 4. The authors attributed these differences to the influence of the alumina support on the properties of ILs which may differ considerably depending on the cation–anion pair.

What should be considered in our case is that the ceramic NF membranes used to cast the SILMs differed significantly from the polymeric ones and the AAOs, as they exhibited tortuosity factors of the order of 140 (eq 4), which is far above the values of 1.4–3.2 usually characterizing the macroporous polymeric supports³⁵ and the value of 1 for AAOs.⁶⁵

Nonetheless, these very specific and highly complex pore structural features have already been taken into account in the prediction of permeability via the definition and use of the κ_g factor. For this reason the large deviations must be primarily attributed to other causes such as the increase of the RTIL viscosity under confinement and the mutual intermolecular blocking effects taking place because of the application of binary CO₂/N₂ mixtures in the experiments. In regard to the latter issue, it is clear (Figure 9 a, b) that as the partial pressure of CO₂ in the binary mixture increases (150–500 mbar) the CO₂/N₂ selectivity of SILM 1 increases, whereas no clear effect is evidenced for SILM 2. The behavior observed for SILM 1 is attributed to the fact that increasing the CO₂ concentration in the feed mixture enhances the probability for the CO₂ molecules to be directly transferred from the gas phase to the pore cavity and absorbed from the confined RTIL phase. In addition, RTIL layers formed on the external surface (outside the pores) absorb more CO₂ which is more efficiently transferred to the pore mouth through surface diffusion.

However, what is more important to note is that the deviation of the experimental CO₂ permeability and CO₂/N₂ selectivity from the predicted one was positive for the SILM 1 ([bmim][TCM]) and negative for the SILM 2 ([emim][TCM]) membrane (Figure 9 a, d). The inexistence of a common trend for both membranes, e.g. for both RTILs,

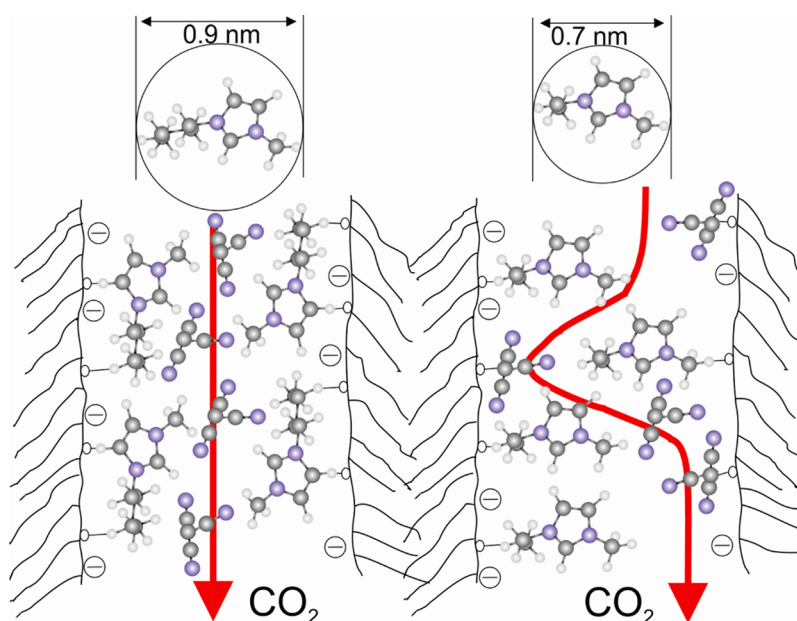


Figure 10. Oriented layering of the [bmim][TCM] compared to the [emim][TCM].

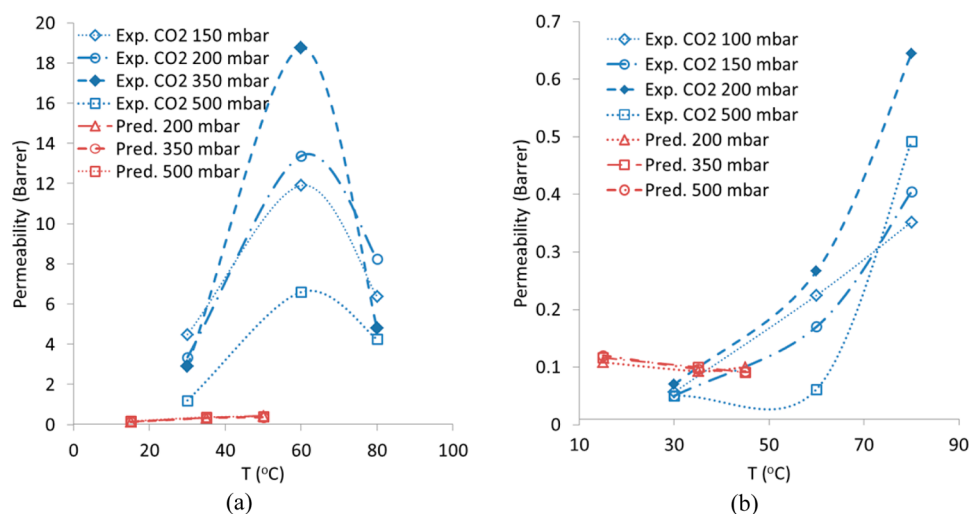


Figure 11. Effect of temperature on the predicted and experimental CO₂ permeability. (a) SILM 1, (b) SILM 2.

implies the effect of additional factors, other than solely the alteration of the RTIL viscosity or the mutual intermolecular blocking effects.

Taking into account that permeability in SILMs scales inversely with viscosity,⁴² our initial expectation was that the prediction would overestimate rather than underestimate the permeability due to a possible increase of the IL viscosity under confinement. This holds for the [emim][TCM], showing that interaction of CO₂ with this IL is of the same nature with this commonly occurring in most of the imidazolium-based ILs, with CO₂ diffusivity strongly depending on the viscosity. On the other hand, as already concluded from the molar volume and temperature dependence of the Henry constants (H_{21}) (section 3.1) and from the temperature dependence of the diffusion factors (D) (section 3.2), the CO₂ interacts strongly with [bmim][TCM]. In this regard, the CO₂ molecules must be gathered at specific sites around the anions into the bulk [bmim][TCM] phase. If these sites are oriented toward the core of the nanopores, then a straight diffusion path is

generated, facilitating the faster transport of CO₂. In fact, infiltration of the bulky ILs into 1 nm pores must result in the formation of a monolayer of IL molecules onto the pore walls. The negatively charged nanoporous silica layer (pzc = 2–4 depending on the nanoparticles size⁶⁶) attracts electrostatically the cations toward the surface of the pore. Due to steric effects, the bulkier [bmim] cations, the longitudinal size of which is close to the pore size,⁶⁷ acquire an optimized geometry and are tilted in respect to the surface, lying with the imidazolium ring almost parallel to the surface plane as shown in Figure 10. In such a conformation the core of the pore is available for hosting the counteranions. It should be noted that a similar orientation of alkyl imidazolium ILs on the silica surface has been observed experimentally with sum-frequency vibrational spectroscopy (SFVS), and the defined tilt of the terminal alkyl group was 25–42°. ⁶⁸ Moreover, density profiles obtained with MD simulations for imidazolium-based ILs confined between parallel silica plates of distances that varied in the range of

4.5–2.5 nm have showed that the interfacial layer defined up to the first density minimum had an excess of cations.⁶⁹

This optimized geometry facilitates gas transport through a hopping mechanism from anion to anion (complexation/decomplexation - diffusion) with the anions located at the central space of the pore, generating a straight diffusion path and leading to permeability values that are higher than the predicted ones. A similar orientation-facilitated transport was observed in a previous work of the authors³⁴ where attachment of the cation on to the nanopore walls was ensured through its silanization and grafting reaction with the surface silanol groups.

On the other hand, the less bulky [emim] cations can be hosted into the pores in a completely random configuration (Figure 10, right), providing space also to the anions to interact with the pore surface. This random deposition generates random diffusion paths that are similar to those occurring in the bulk, unconfined, IL phase. As expected, due to the higher viscosity of the [emim][TCM] under confinement, the experimental permeability values for SILM 2 were lower than the predicted.

3.6.1. Effect of Temperature on the Experimental Permeability. The different gas diffusion mechanism occurring in membranes SILM 1 and SILM 2 is also evidenced by the completely different effect of temperature on the experimentally derived CO₂ permeability. It can be seen (Figure 11a) that the CO₂ permeability in SILM 1 levels off at the temperature of 60 °C, whereas that of SILM 2 scales proportionally with temperature, following the general rule of permeability dependence on decreasing viscosity while a sudden increase is observed after 60 °C. Considering again the possibility to correlate the RTIL viscosity with temperature via the Andrade equation,⁴⁹ we have further applied an Arrhenius type of analysis to define the temperature dependence of the permeability for SILM 2. The analysis produced a very high activation energy value of $E_a = 37 \text{ kJ mol}^{-1}$ when the respective analysis for the diffusion coefficients (D) obtained from the transient absorption experiments of the relevant RTIL [emim][TCM] was of the order of 7.33 kJ mol^{-1} . Particularly, if the [emim][TCM] had retained its liquid state after its confinement into the nanopores of SILM 2, then, as shown in Figure 11b, the CO₂ permeabilities for SILM 2 should scale inversely with temperature (predicted permeability) because of the opposite effect of temperature on D and S . This enhanced dependence of permeability on temperature, especially beyond the value of 60 °C, leads to the conclusion that [emim][TCM] must have suffered a liquid–solid transition under confinement into the nanopores at room temperature. To further evidence this, we have performed DSC runs in a small slice of the selective layer of SILM 2, in the bulk [bmim][TCM], and in two other SILMs casted with [emim][TCM] on ceramic membranes with selective layer pore sizes of 5 and 10 nm. The samples were quenched down to –180 °C and then heated to the temperature of 150 °C with a rate of 10 °C min^{-1} . The thermograph of the bulk [bmim][TCM] (Supporting Information) showed only the T_g transition occurring at –92.3 °C but no further peaks corresponding to crystallization and melting. On the other hand, the thermographs of the SILM 2 and the SILMs developed on the 5 and 10 nm supports (Supporting Information) showed a shift of the T_g transitions to the temperatures of –100.7 °C, –96.5 °C, and –100.15 °C, respectively. Moreover, the SILMs casted on the 5 and 10 nm supports gave very intense exotherms (crystallization) at –50

°C and –40 °C, followed by intense endotherms at –32 °C and –27 °C, respectively, the latter corresponding to melting of the IL confined into the 5 and 10 nm pores. Contrary, the thermograph of the SILM 2, after the T_g transition, contained solely a very intense and broad endotherm at 73.7 °C, which corresponded to melting of the IL confined into the 1 nm pores. It is important to note that no occurrence of crystallization was found for SILM 2 upon heating (after the quenching), something denoting that [emim][TCM] was already crystalline at room temperature after its confinement into the nanopores. This phase transition explains the negative deviation of the experimental permeability from the prediction at room temperature for SILM 2 (Figure 9 a). Moreover, the melting at 73.7 °C coincides with the sudden upraise of the permeability from 60 to 80 °C (Figure 11b).

In regard to the diffusion mechanism in SILM 1, the level off occurring on the CO₂ permeability vs temperature at 60 °C (Figure 11a) was something expected. Practically, it indicates the opposite effect of temperature increment to the solubility (S) and diffusivity (D) of CO₂ in the RTIL; e.g., as the temperature increases the solubility decreases and the diffusivity increases up to the point that their product (e.g., permeability) levels off. However, we should emphasize how much more intense the effect of temperature is on the experimental permeability compared to the predicted one (Figure 11a). This implies that the diffusion mechanism through the confined ionic liquid phase differs significantly from this occurring through a bulk IL casted in a macroporous support. In our case (nanoconfinement), activated phenomena resembling these occurring in micropores, such as the transfer and absorption of CO₂ directly from the gas phase on the IL phase confined inside the nanopores, the surface diffusion of CO₂ molecules through the externally deposited IL toward the pore mouth, and their further transfer to the IL phase under confinement and the hopping mechanism from active site to active site as described in Figure 10 (left), seem to be much more intense; e.g., the energy required for overcoming all the above-mentioned barriers is enhanced.

4. CONCLUSIONS

A detailed methodology for the prediction of the gas permeability in supported ionic liquid membranes (SILMs) was developed. The method revealed considerable deviations of the predictions from the experimental results obtained with SILMs casted from 1-ethyl-3-methylimidazolium ([emim]-[TCM]) and 1-butyl-3-methylimidazolium tricyanomethanide into nanofiltration (NF) membranes. The deviations were not attributed to the complex pore structural characteristics of the NF membranes but rather to the specific layering phenomena and crystallization of the IL phase occurring under extreme confinement into the nanopores. Specifically, bulkier cations such as the [bmim], with longitudinal molecular size almost equal to the pore opening size, obtain a specific orientation and are tilted with respect to the pore surface. This optimized orientation leaves the core of the pore available to be occupied by the counteranions [TCM]. In this way, straight diffusion paths are generated that facilitate the transport of CO₂ through a hopping mechanism from anion to anion. The as-developed membranes exhibited enhanced CO₂ permeability factors combined with significant CO₂/N₂ separation capacity. On the other hand, the [emim][TCM] underwent a phase transition from liquid to solid under extreme confinement into the nanopores. The CO₂ diffusion through the solidified IL

became much slower than predicted, whereas the CO₂/N₂ separation capacity remained high. As a general conclusion empirical models that predict accurately the permeability performance of SILMs based on the physicochemical properties of the relevant RTILs cannot be applied in the case of ILs confinement into nanopores.

■ ASSOCIATED CONTENT

■ Supporting Information

The physical properties of the new ionic liquids, the experimental setup and equations used to derive the permeability and selectivity factors, the assumptions made for deriving the appropriate solution of the transient absorption equation, and the experimental transient absorption curves fitted with the solution as well as the differential scanning calorimetry thermographs of the SILM 2 and of the SILMs developed on 5 and 10 nm membranes. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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